

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12405030
Kind Code	B2
Date of Patent	September 02, 2025
Inventor(s)	Cabrera; Robert Edward et al.

Top fired outdoor gas heat exchanger

Abstract

A furnace for a heating, ventilation, and air conditioning (HVAC) unit includes a heat exchange tube configured to flow combustion products therethrough and place the combustion products in a heat exchange relationship with an air flow directed across the heat exchange tube. The furnace also includes a burner assembly fluidly coupled to a first port of the heat exchange tube and configured to generate the combustion products directed into the heat exchange tube via the first port, and a draft inducer blower fluidly coupled to a second port of the heat exchange tube and configured to draw the combustion products through the heat exchange tube. The burner assembly is higher in position than the draft inducer blower relative to a base of the HVAC unit.

Inventors: Cabrera; Robert Edward (Wichita, KS), Jayarathne; Madhuka Manuranga (Wichita, KS)

Applicant: Johnson Controls Tyco IP Holdings LLP (Milwaukee, WI)

Family ID: 86384539

Assignee: Tyco Fire & Security GmbH (Neuhausen am Rheinfall, CH)

Appl. No.: 17/534192

Filed: November 23, 2021

Prior Publication Data

Document Identifier	Publication Date
US 20230160605 A1	May. 25, 2023

Publication Classification

Int. Cl.: F24H8/00 (20220101); F24H3/08 (20220101); F24H9/1881 (20220101)

U.S. Cl.:

Field of Classification Search

USPC: None

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
5775318	12/1997	Haydock	126/110R	F28F 9/26
5839505	12/1997	Ludwig	165/DIG.497	F28F 1/06
6352431	12/2001	Gatley, Jr.	126/110A	F27D 9/00
8056553	12/2010	Khan	126/110R	F24H 3/087
9709341	12/2016	Naruse	N/A	F28D 1/0477
10584896	12/2019	Wilson et al.	N/A	N/A
10753687	12/2019	Plummer	N/A	F28D 7/087
2002/0005275	12/2001	O'Donnell	165/163	F24H 9/0026
2005/0161209	12/2004	Havard, Jr.	165/177	F24H 3/087
2008/0029243	12/2007	O'Donnell	165/184	F28F 1/06
2022/0042716	12/2021	Kanagala	N/A	F28D 7/06
2023/0330451	12/2022	Jones	N/A	A62B 23/02

Primary Examiner: Maroney; Jenna M

Attorney, Agent or Firm: Fletcher Yoder, PC

Background/Summary

BACKGROUND

(1) This section is intended to introduce the reader to various aspects of art that may be related to various aspects of the present disclosure and are described below. This discussion is believed to be helpful in providing the reader with background information to facilitate a better understanding of the various aspects of the present disclosure. Accordingly, it should be noted that these statements are to be read in this light, and not as admissions of prior art.

(2) Heating, ventilation, and air conditioning (HVAC) systems are utilized to control environmental properties, such as temperature and humidity, for occupants of residential, commercial, and industrial environments. The HVAC systems may control the environmental properties through control of an air flow delivered to the environment. For example, an HVAC system may include several heat exchangers, such as a heat exchanger configured to place an air flow in a heat exchange relationship with a refrigerant of a vapor compression circuit (e.g., evaporator, condenser), a heat exchanger configured to place an air flow in a heat exchange relationship with combustion products (e.g., a furnace), or both. In general, the heat exchange relationship(s) may cause a change in pressures and/or temperatures of the air, the refrigerant, the combustion products, or any combination thereof. As the temperatures and/or pressures of the above-described fluids change, liquid condensate may be formed in or on the associated heat exchangers.

SUMMARY

(3) A summary of certain embodiments disclosed herein is set forth below. It should be noted that these aspects are presented merely to provide the reader with a brief summary of these certain embodiments and that these aspects are not intended to limit the scope of this disclosure. Indeed, this disclosure may encompass a variety of aspects that may not be set forth below.

(4) In an embodiment, a furnace for a heating, ventilation, and air conditioning (HVAC) unit includes a heat exchange tube configured to flow combustion products therethrough and place the combustion products in a heat exchange relationship with an air flow directed across the heat exchange tube. The furnace also includes a burner assembly fluidly coupled to a first port of the heat exchange tube and configured to generate the combustion products directed into the heat exchange tube via the first port, and a draft inducer blower fluidly coupled to a second port of the heat exchange tube and configured to draw the combustion products through the heat exchange tube. The burner assembly is higher in position than the draft inducer blower relative to a base of the HVAC unit.

(5) In another embodiment, a furnace for a heating, ventilation, and air conditioning (HVAC) system includes a panel comprising an inlet and an outlet, and a heat exchange tube fluidly coupled to the inlet and to the outlet on a first side of the panel. The heat exchange tube is configured to direct combustion products from the inlet to the outlet and place the combustion products in a heat exchange relationship with an air flow directed across the heat exchange tube along an air flow path through the furnace. The furnace also includes a burner assembly coupled to a second side of the panel at a first position along a vertical axis, and a draft inducer blower coupled to the second side of the panel at a second position along the vertical axis. The first position is above the second position along the vertical axis. The burner assembly is configured to generate the combustion products and direct the combustion products into the heat exchange tube via the inlet, the draft inducer blower is configured to draw the combustion products through the heat exchange tube towards the outlet.

(6) In another embodiment, a furnace for a heating, ventilation, and air conditioning (HVAC) system includes a heat exchange tube having a first port configured to receive combustion products and a second port configured to discharge the combustion products. The heat exchange tube is configured to direct the combustion products from the first port to the second port. The furnace also includes a burner assembly fluidly coupled to the first port, and a draft inducer blower fluidly coupled to the second port. The burner assembly is configured to generate the combustion products and direct the combustion products into the heat exchange tube via the first port, and the draft inducer blower is configured to draw the combustion products through the heat exchange tube and remove the combustion products from the heat exchange tube via the second port. The first port is above the second port relative to gravity, and the heat exchange tube is configured to discharge liquid condensate from the heat exchange tube via the second port.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) Various aspects of this disclosure may be better understood upon reading the following detailed description and upon reference to the drawings in which:

(2) FIG. 1 is a perspective view of a building having an embodiment of a heating, ventilation, and air conditioning (HVAC) system for environmental management that may employ one or more HVAC units, in accordance with an aspect of the present disclosure;

(3) FIG. 2 is a perspective view of an embodiment of a packaged HVAC unit that may be used in the HVAC system of FIG. 1, in accordance with an aspect of the present disclosure;

(4) FIG. 3 is a cutaway perspective view of an embodiment of a residential, split HVAC system, in accordance with an aspect of the present disclosure;

- (5) FIG. 4 is a schematic illustration of an embodiment of a vapor compression system that can be used in any of the systems of FIGS. 1-3, in accordance with an aspect of the present disclosure;
- (6) FIG. 5 is a perspective view of an embodiment of an HVAC unit, in accordance with an aspect of the present disclosure;
- (7) FIG. 6 is a side view of an embodiment of a furnace, in accordance with an aspect of the present disclosure;
- (8) FIG. 7 is a schematic side view of an embodiment of a furnace, illustrating flow of liquid condensate within the furnace, in accordance with an aspect of the present disclosure;
- (9) FIG. 8 is an schematic side view of an embodiment of a draft inducer of a furnace, in accordance with an aspect of the present disclosure; and
- (10) FIG. 9 is a front perspective view of an embodiment of a furnace, in accordance with an aspect of the present disclosure.
- (11) FIG. 10 is a front perspective view of an embodiment of a furnace, in accordance with an aspect of the present disclosure.

DETAILED DESCRIPTION

(12) One or more specific embodiments will be described below. In an effort to provide a concise description of these embodiments, not all features of an actual implementation are described in the specification. It should be noted that in the development of any such actual implementation, as in any engineering or design project, numerous implementation-specific decisions must be made to achieve the developers' specific goals, such as compliance with system-related and business-related constraints, which may vary from one implementation to another. Moreover, it should be noted that such a development effort might be complex and time consuming, but would nevertheless be a routine undertaking of design, fabrication, and manufacture for those of ordinary skill having the benefit of this disclosure.

(13) When introducing elements of various embodiments of the present disclosure, the articles “a,” “an,” and “the” are intended to mean that there are one or more of the elements. The terms “comprising,” “including,” and “having” are intended to be inclusive and mean that there may be additional elements other than the listed elements. Additionally, it should be noted that references to “one embodiment” or “an embodiment” of the present disclosure are not intended to be interpreted as excluding the existence of additional embodiments that also incorporate the recited features.

(14) The present disclosure is directed to a heat exchanger for heating, ventilation, and air conditioning (HVAC) systems configured to increase the temperature of an air flow directed through the HVAC system. In some embodiments, the heat exchanger (e.g., furnace) may be disposed in a packaged outdoor unit or a rooftop unit configured to both heat and cool an air flow, such as a supply air flow that is conditioned and directed to a conditioned space (e.g., a building). For example, the furnace may include a heat exchanger having tubes that is configured to receive relatively hot combustion products (e.g., flue gas) generated via a burner assembly. The furnace may also include a draft inducer (e.g., draft inducer blower) configured to circulate the combustion products through the tubes of the heat exchanger. Further, the furnace may include a blower configured to direct the supply air flow across the tubes, thereby placing the supply air flow in a heat exchange relationship with the relatively hot combustion products to heat the supply air flow.

(15) In some circumstances, liquid condensate may form in or on the above-described heat exchanger. For example, during a cooling mode of the HVAC system (e.g., when the furnace is in an inoperative mode or shut-off), relatively cool supply air flow may be directed across the tubes of the heat exchanger. The relatively cool supply air flow may cause air within the tubes of the heat exchanger (e.g., ambient air) to cool, thereby causing moisture contained within the air to condense. As the air within the tubes condenses, liquid condensate may form within the tubes. However, collection of condensate within the tubes may adversely affect the heat exchanger, and therefore it may be desirable to drain the condensate from the heat exchanger. Unfortunately,

traditional heat exchangers (e.g., furnaces) may be configured in a manner that does not adequately allow the condensate to drain from the heat exchanger. For example, existing designs may cause condensate to flow via gravity to the burner assembly, which may lead to degradation, operating interruptions, and/or inefficiencies in the heat exchanger. That is, traditional heat exchanger configurations typically include a burner assembly connected to heat exchange tubes at a base (e.g., bottom side, near a drain outlet) of the heat exchanger and a draft inducer connected to the heat exchange tubes near a top side of the heat exchanger. In such a configuration, the burner assembly is susceptible to potential degradation from liquid or liquid condensate that may flow toward the burner assembly via gravity.

(16) It is now recognized that improved heat exchanger configurations and related features are desired to limit an amount of liquid condensate that may reach the burner assembly, thereby limiting potential degradation and inefficiencies of a furnace. In accordance with the present techniques, the heat exchanger may be configured to enable a liquid (e.g., condensate) within the heat exchange tubes to flow towards a drain outlet at a base of the heat exchanger. For example, one or more segments of the tubes may be positioned at an angle relative to horizontal to enable drainage of liquid therein via gravity. A draft inducer may be fluidly connected to the heat exchange tubes at a base of the heat exchanger and proximate to the drain outlet of the heat exchanger. A burner assembly may also be fluidly connected to the heat exchange tubes at a position above (e.g., top-fired heat exchanger) the draft inducer relative to gravity (e.g., near the top of the heat exchanger), such that liquid condensate formed within the heat exchange tubes (e.g., via condensation) will be directed away from the burner assembly and towards the drain outlet via gravity. The term “top-fired heat exchanger” used herein may refer to a general configuration in which the burner assembly is connected to a first end or port of the heat exchange tubes at a first position, the draft inducer is connected to a second end or port of the heat exchange tubes at a second position, and the first position of the burner assembly is higher than the second position of the draft inducer, relative to gravity. Such a configuration may limit an amount of liquid condensate from reaching the burner assembly, thereby increasing efficiency and reducing a likelihood of degradation to certain aspects of the furnace.

(17) As will be appreciated, the heat exchanger systems disclosed herein may be used in association with any of a variety of HVAC systems, including those in residential and commercial settings. For example, the heat exchanger systems may be utilized in a rooftop unit (RTU), a dedicated outdoor air system, or a split system. Non-limiting examples of systems that may use the heat exchanger system of the present disclosure are described herein with respect to FIGS. 1-4.

(18) Turning now to the drawings, FIG. 1 illustrates a heating, ventilation, and air conditioning (HVAC) system for building environmental management that may employ one or more HVAC units. As used herein, an HVAC system includes any number of components configured to enable regulation of parameters related to climate characteristics, such as temperature, humidity, air flow, pressure, air quality, and so forth. For example, an “HVAC system” as used herein is defined as conventionally understood and as further described herein. Components or parts of an “HVAC system” may include, but are not limited to, all, some of, or individual parts such as a heat exchanger, a heater, an air flow control device, such as a fan, a sensor configured to detect a climate characteristic or operating parameter, a filter, a control device configured to regulate operation of an HVAC system component, a component configured to enable regulation of climate characteristics, or a combination thereof. An “HVAC system” is a system configured to provide such functions as heating, cooling, ventilation, dehumidification, pressurization, refrigeration, filtration, or any combination thereof. The embodiments described herein may be utilized in a variety of applications to control climate characteristics, such as residential, commercial, industrial, transportation, or other applications where climate control is desired.

(19) In the illustrated embodiment, a building **10** is air conditioned by a system that includes an HVAC unit **12**. The building **10** may be a commercial structure or a residential structure. As shown,

the HVAC unit **12** is disposed on the roof of the building **10**; however, the HVAC unit **12** may be located in other equipment rooms or areas adjacent the building **10**. The HVAC unit **12** may be a single package unit containing other equipment, such as a blower, integrated air handler, and/or auxiliary heating unit. In other embodiments, the HVAC unit **12** may be part of a split HVAC system, such as the system shown in FIG. **3**, which includes an outdoor HVAC unit **58** and an indoor HVAC unit **56**.

(20) The HVAC unit **12** is an air cooled device that implements a refrigeration cycle to provide conditioned air to the building **10**. Specifically, the HVAC unit **12** may include one or more heat exchangers across which an air flow is passed to condition the air flow before the air flow is supplied to the building. In the illustrated embodiment, the HVAC unit **12** is a rooftop unit (RTU) that conditions a supply air stream, such as environmental air and/or a return air flow from the building **10**. After the HVAC unit **12** conditions the air, the air is supplied to the building **10** via ductwork **14** extending throughout the building **10** from the HVAC unit **12**. In certain embodiments, the HVAC unit **12** may be a heat pump that provides both heating and cooling to the building with one refrigeration circuit configured to operate in different modes. In other embodiments, the HVAC unit **12** may include one or more refrigeration circuits for cooling an air stream and a furnace for heating the air stream.

(21) A control device **16**, one type of which may be a thermostat, may be used to designate the temperature of the conditioned air. The control device **16** also may be used to control the flow of air through the ductwork **14**. For example, the control device **16** may be used to regulate operation of one or more components of the HVAC unit **12** or other components, such as dampers and fans, within the building **10** that may control flow of air through and/or from the ductwork **14**. In some embodiments, other devices may be included in the system, such as pressure and/or temperature transducers or switches that sense the temperatures and pressures of the supply air, return air, and so forth. Moreover, the control device **16** may include computer systems that are integrated with or separate from other building control or monitoring systems, and even systems that are remote from the building **10**.

(22) FIG. **2** is a perspective view of an embodiment of the HVAC unit **12**. In the illustrated embodiment, the HVAC unit **12** is a single package unit that may include one or more independent refrigeration circuits and components that are tested, charged, wired, piped, and ready for installation. The HVAC unit **12** may provide a variety of heating and/or cooling functions, such as cooling only, heating only, cooling with electric heat, cooling with dehumidification, cooling with gas heat, or cooling with a heat pump. As described above, the HVAC unit **12** may directly cool and/or heat an air stream provided to the building **10** to condition a space in the building **10**.

(23) As shown in the illustrated embodiment of FIG. **2**, a cabinet **24** encloses the HVAC unit **12** and provides structural support and protection to the internal components from environmental and other contaminants. In some embodiments, the cabinet **24** may be constructed of galvanized steel and insulated with aluminum foil faced insulation. Rails **26** may be joined to the bottom perimeter of the cabinet **24** and provide a foundation for the HVAC unit **12**. In certain embodiments, the rails **26** may provide access for a forklift and/or overhead rigging to facilitate installation and/or removal of the HVAC unit **12**. In some embodiments, the rails **26** may fit onto “curbs” on the roof to enable the HVAC unit **12** to provide air to the ductwork **14** from the bottom of the HVAC unit **12** while blocking elements such as rain from leaking into the building **10**.

(24) The HVAC unit **12** includes heat exchangers **28** and **30** in fluid communication with one or more refrigeration circuits. Tubes within the heat exchangers **28** and **30** may circulate refrigerant, such as R-410A, through the heat exchangers **28** and **30**. The tubes may be of various types, such as multichannel tubes, conventional copper or aluminum tubing, and so forth. Together, the heat exchangers **28** and **30** may implement a thermal cycle in which the refrigerant undergoes phase changes and/or temperature changes as it flows through the heat exchangers **28** and **30** to produce heated and/or cooled air. For example, the heat exchanger **28** may function as a condenser where

heat is released from the refrigerant to ambient air, and the heat exchanger **30** may function as an evaporator where the refrigerant absorbs heat to cool an air stream. In other embodiments, the HVAC unit **12** may operate in a heat pump mode where the roles of the heat exchangers **28** and **30** may be reversed. That is, the heat exchanger **28** may function as an evaporator and the heat exchanger **30** may function as a condenser. In further embodiments, the HVAC unit **12** may include a furnace for heating the air stream that is supplied to the building **10**. While the illustrated embodiment of FIG. 2 shows the HVAC unit **12** having two of the heat exchangers **28** and **30**, in other embodiments, the HVAC unit **12** may include one heat exchanger or more than two heat exchangers.

(25) The heat exchanger **30** is located within a compartment **31** that separates the heat exchanger **30** from the heat exchanger **28**. Fans **32** draw air from the environment through the heat exchanger **28**. Air may be heated and/or cooled as the air flows through the heat exchanger **28** before being released back to the environment surrounding the HVAC unit **12**. A blower assembly **34**, powered by a motor **36**, draws air through the heat exchanger **30** to heat or cool the air. The heated or cooled air may be directed to the building **10** by the ductwork **14**, which may be connected to the HVAC unit **12**. Before flowing through the heat exchanger **30**, the conditioned air flows through one or more filters **38** that may remove particulates and contaminants from the air. In certain embodiments, the filters **38** may be disposed on the air intake side of the heat exchanger **30** to prevent contaminants from contacting the heat exchanger **30**.

(26) The HVAC unit **12** also may include other equipment for implementing the thermal cycle. Compressors **42** increase the pressure and temperature of the refrigerant before the refrigerant enters the heat exchanger **28**. The compressors **42** may be any suitable type of compressors, such as scroll compressors, rotary compressors, screw compressors, or reciprocating compressors. In some embodiments, the compressors **42** may include a pair of hermetic direct drive compressors arranged in a dual stage configuration **44**. However, in other embodiments, any number of the compressors **42** may be provided to achieve various stages of heating and/or cooling. Additional equipment and devices may be included in the HVAC unit **12**, such as a solid-core filter drier, a drain pan, a disconnect switch, an economizer, pressure switches, phase monitors, and humidity sensors, among other things.

(27) The HVAC unit **12** may receive power through a terminal block **46**. For example, a high voltage power source may be connected to the terminal block **46** to power the equipment. The operation of the HVAC unit **12** may be governed or regulated by a control board **48**. The control board **48** may include control circuitry connected to a thermostat, sensors, and alarms. One or more of these components may be referred to herein separately or collectively as the control device **16**. The control circuitry may be configured to control operation of the equipment, provide alarms, and monitor safety switches. Wiring **49** may connect the control board **48** and the terminal block **46** to the equipment of the HVAC unit **12**.

(28) FIG. 3 illustrates a residential heating and cooling system **50**, also in accordance with present techniques. The residential heating and cooling system **50** may provide heated and cooled air to a residential structure, as well as provide outside air for ventilation and provide improved indoor air quality (IAQ) through devices such as ultraviolet lights and air filters. In the illustrated embodiment, the residential heating and cooling system **50** is a split HVAC system. In general, a residence **52** conditioned by a split HVAC system may include refrigerant conduits **54** that operatively couple the indoor unit **56** to the outdoor unit **58**. The indoor unit **56** may be positioned in a utility room, an attic, a basement, and so forth. The outdoor unit **58** is typically situated adjacent to a side of residence **52** and is covered by a shroud to protect the system components and to prevent leaves and other debris or contaminants from entering the unit. The refrigerant conduits **54** transfer refrigerant between the indoor unit **56** and the outdoor unit **58**, typically transferring primarily liquid refrigerant in one direction and primarily vaporized refrigerant in an opposite direction.

(29) When the system shown in FIG. 3 is operating as an air conditioner, a heat exchanger 60 in the outdoor unit 58 serves as a condenser for re-condensing vaporized refrigerant flowing from the indoor unit 56 to the outdoor unit 58 via one of the refrigerant conduits 54. In these applications, a heat exchanger 62 of the indoor unit functions as an evaporator. Specifically, the heat exchanger 62 receives liquid refrigerant, which may be expanded by an expansion device, and evaporates the refrigerant before returning it to the outdoor unit 58.

(30) The outdoor unit 58 draws environmental air through the heat exchanger 60 using a fan 64 and expels the air above the outdoor unit 58. When operating as an air conditioner, the air is heated by the heat exchanger 60 within the outdoor unit 58 and exits the unit at a temperature higher than it entered. The indoor unit 56 includes a blower or fan 66 that directs air through or across the indoor heat exchanger 62, where the air is cooled when the system is operating in air conditioning mode. Thereafter, the air is passed through ductwork 68 that directs the air to the residence 52. The overall system operates to maintain a desired temperature as set by a system controller. When the temperature sensed inside the residence 52 is higher than the set point on the thermostat, or the set point plus a small amount, the residential heating and cooling system 50 may become operative to refrigerate additional air for circulation through the residence 52. When the temperature reaches the set point, or the set point minus a small amount, the residential heating and cooling system 50 may stop the refrigeration cycle temporarily.

(31) The residential heating and cooling system 50 may also operate as a heat pump. When operating as a heat pump, the roles of heat exchangers 60 and 62 are reversed. That is, the heat exchanger 60 of the outdoor unit 58 will serve as an evaporator to evaporate refrigerant and thereby cool air entering the outdoor unit 58 as the air passes over the outdoor heat exchanger 60. The indoor heat exchanger 62 will receive a stream of air blown over it and will heat the air by condensing the refrigerant.

(32) In some embodiments, the indoor unit 56 may include a furnace system 70. For example, the indoor unit 56 may include the furnace system 70 when the residential heating and cooling system 50 is not configured to operate as a heat pump. The furnace system 70 may include a burner assembly and heat exchanger, among other components, inside the indoor unit 56. Fuel is provided to the burner assembly of the furnace system 70 where it is mixed with air and combusted to form combustion products. The combustion products may pass through tubes or piping in a heat exchanger, separate from heat exchanger 62, such that air directed by the blower or fan 66 passes over the tubes or pipes and extracts heat from the combustion products. The heated air may then be routed from the furnace system 70 to the ductwork 68 for heating the residence 52.

(33) FIG. 4 is an embodiment of a vapor compression system 72 that can be used in any of the systems described above. The vapor compression system 72 may circulate a refrigerant through a circuit starting with a compressor 74. The circuit may also include a condenser 76, an expansion valve(s) or device(s) 78, and an evaporator 80. The vapor compression system 72 may further include a control panel 82 that has an analog to digital (A/D) converter 84, a microprocessor 86, a non-volatile memory 88, and/or an interface board 90. The control panel 82 and its components may function to regulate operation of the vapor compression system 72 based on feedback from an operator, from sensors of the vapor compression system 72 that detect operating conditions, and so forth.

(34) In some embodiments, the vapor compression system 72 may use one or more of a variable speed drive (VSDs) 92, a motor 94, the compressor 74, the condenser 76, the expansion valve or device 78, and/or the evaporator 80. The motor 94 may drive the compressor 74 and may be powered by the variable speed drive (VSD) 92. The VSD 92 receives alternating current (AC) power having a particular fixed line voltage and fixed line frequency from an AC power source, and provides power having a variable voltage and frequency to the motor 94. In other embodiments, the motor 94 may be powered directly from an AC or direct current (DC) power source. The motor 94 may include any type of electric motor that can be powered by a VSD or

directly from an AC or DC power source, such as a switched reluctance motor, an induction motor, an electronically commutated permanent magnet motor, or another suitable motor.

(35) The compressor **74** compresses a refrigerant vapor and delivers the vapor to the condenser **76** through a discharge passage. In some embodiments, the compressor **74** may be a centrifugal compressor. The refrigerant vapor delivered by the compressor **74** to the condenser **76** may transfer heat to a fluid passing across the condenser **76**, such as ambient or environmental air **96**. The refrigerant vapor may condense to a refrigerant liquid in the condenser **76** as a result of thermal heat transfer with the environmental air **96**. The liquid refrigerant from the condenser **76** may flow through the expansion device **78** to the evaporator **80**.

(36) The liquid refrigerant delivered to the evaporator **80** may absorb heat from another air stream, such as a supply air stream **98** provided to the building **10** or the residence **52**. For example, the supply air stream **98** may include ambient or environmental air, return air from a building, or a combination of the two. The liquid refrigerant in the evaporator **80** may undergo a phase change from the liquid refrigerant to a refrigerant vapor. In this manner, the evaporator **80** may reduce the temperature of the supply air stream **98** via thermal heat transfer with the refrigerant. Thereafter, the vapor refrigerant exits the evaporator **80** and returns to the compressor **74** by a suction line to complete the cycle.

(37) In some embodiments, the vapor compression system **72** may further include a reheat coil in addition to the evaporator **80**. For example, the reheat coil may be positioned downstream of the evaporator relative to the supply air stream **98** and may reheat the supply air stream **98** when the supply air stream **98** is overcooled to remove humidity from the supply air stream **98** before the supply air stream **98** is directed to the building **10** or the residence **52**.

(38) It should be appreciated that any of the features described herein may be incorporated with the HVAC unit **12**, the residential heating and cooling system **50**, or other HVAC systems.

Additionally, while the features disclosed herein are described in the context of embodiments that directly heat and cool a supply air stream provided to a building or other load, embodiments of the present disclosure may be applicable to other HVAC systems as well. For example, the features described herein may be applied to mechanical cooling systems, free cooling systems, chiller systems, or other heat pump or refrigeration applications.

(39) Further, any of the systems illustrated in FIGS. **1-4** may include or operate in conjunction with a furnace in accordance with the present disclosure, such as the furnace system **70** of FIG. **3**. For example, the furnace system **70** of FIG. **3** may generate combustion products, sometimes referred to as flue gas or exhaust gas, and then rout the combustion products through tubes (or coils) of the furnace system **70**. During an operative mode (e.g., heating mode), a supply air flow may be forced across the tubes of the furnace system **70**, for example by a fan or blower, such that the supply air flow is heated by the combustion products in the tubes of the furnace system **70** prior to delivery of the heated air flow to a conditioned space. Similarly, during a cooling mode (e.g., when the furnace is shut-off or inoperative), ambient or other air may remain in the tubes of the furnace, and a relatively cool supply air flow may be directed across the tubes. As the air within the tubes is cooled via heat exchange with the supply air flow, liquid condensate may form inside of the tubes of the furnace system **70**.

(40) In accordance with the present disclosure, a heat exchanger (e.g., a furnace) may be coupled to a heat source, such as a burner assembly (e.g., burner) that generates combustion products, to provide heat to a supply air flow directed across the heat exchanger via a supply air source (e.g., blower, fan). The heat exchanger may also be coupled to a draft inducer that directs (e.g., draws) the combustion products through one or more heat exchange tubes of the heat exchanger. The burner assembly may be fluidly connected to a first port of the heat exchange tubes at a first position proximate a top portion of the heat exchanger, and the draft inducer may be fluidly connected to a second port of the heat exchange tubes at a second position near a base portion of the heat exchanger. A drain outlet may also be located near the second end of the heat exchange

tubes and may be configured to drain liquid condensate that forms within the heat exchange tubes during certain operations of the HVAC system, as described above. The first position (e.g., position of the burner assembly) may be higher relative to gravity than the second position (e.g., position of the draft inducer), thereby resulting in a top-fired heat exchanger configuration. By positioning the burner assembly at or near the top of the heat exchanger, liquid condensate formed within the heat exchange tubes may be directed away from the burner assembly at the first position and may instead be directed toward the drain outlet at the second position via the draft inducer and via gravity. In this manner, heat exchangers having the configuration discussed herein may be less susceptible to degradation, operating interruptions, and/or inefficiencies that may otherwise occur in traditional heat exchangers.

(41) With this in mind, FIG. 5 is a perspective view of an embodiment of a packaged HVAC unit **100** that may employ one or more of the heat exchangers disclosed herein. In the illustrated embodiment, the packaged HVAC unit **100** includes multiple components enclosed within an internal volume of a housing **102** of the packaged HVAC unit **100**. The packaged HVAC unit **100** may be configured to circulate air and therefore may include a return section **104** to receive an air flow, such as a return air flow from the building **10**, and a supply section **106** to output an air flow, such as a supply air flow. As an example, the packaged HVAC unit **100** may be located in an outside environment, such as on a rooftop, and may be coupled to ductwork that directs air to and/or from rooms or other areas within a building, such as the building **10** of FIG. 1. The ductwork may couple to the return section **104** and the supply section **106**. In this manner, the packaged HVAC unit **100** may circulate air in the building **10**.

(42) In addition to circulating air, the packaged HVAC unit **100** may change the temperature of the supply air flow directed therethrough. For example, the packaged HVAC unit **100** may include a refrigerant circuit that circulates a refrigerant therethrough, where the refrigerant circuit is in thermal communication with the air flow. The refrigerant may flow through a condenser **108**, where the refrigerant may be cooled. FIG. 5 illustrates the condenser **108** as including a fan that may direct ambient air across the condenser **108** to remove heat from the refrigerant via convection, but in other embodiments, the condenser **108** may use another means of cooling the refrigerant, such as via a coolant. After being cooled, the refrigerant may then flow through an evaporator **110**, where the refrigerant may absorb heat from the air flow (e.g., supply air flow) directed across the evaporator **110**. Thus, the refrigerant may be heated, and the air flow may be cooled at the evaporator **110**. After being heated at the evaporator **110**, the refrigerant may return to the condenser **108** where it may once again be cooled. It should be appreciated that the refrigerant circuit may include other components, such as a compressor, expansion valve, and so forth, that enable conditioning of the supply air flow via the refrigerant.

(43) The packaged HVAC unit **100** may also be configured to operate in a heating mode and a cooling mode. During operation of the heating mode, air may be received by the packaged HVAC unit **100** at the return section **104** to enter an air flow path. As mentioned, air (e.g., return air) may be received from ductwork that is connected to a building. However, in other embodiments, air received by the packaged HVAC unit **100** may be ambient air, such as from an outside environment. In certain embodiments, the supply air flow directed through the packaged HVAC unit **100** may include air from the return section **104** as well as ambient air. After the air flow enters the packaged HVAC unit **100**, the air flow may pass across a filter **112**. The filter **112** may remove particles from the air flow, such as dirt or other debris. The filter **112** may be a pleated filter, an electrostatic filter, a HEPA filter, or a fiber glass filter that traps the debris when the air flow passes through the filter **112**. After being filtered, the air flow may be directed to the evaporator **110**. As discussed above, at the evaporator **110**, the air flow may be cooled by transferring heat to the refrigerant within the evaporator **110**. In addition, cooling the air flow may also remove moisture from the air flow and thus, the packaged HVAC unit **100** may also dehumidify the air flow. Once cooled, the air flow may be directed through a blower **114**, which may increase the velocity of the

air flow and discharge the air flow as supply air via the supply section **106** of the packaged HVAC unit **100**. Thereafter, the supply air flow may be circulated through the ductwork. In some embodiments, the blower **114** may also operate to draw air through the return section **104** and thereby function to both draw in and expel air.

(44) In some modes of operation (e.g., a heating mode), prior to exiting the packaged HVAC unit **100**, the air may be heated by a heat exchanger **116** (e.g., a furnace). By way of example, the heat exchanger **116** may be coupled to a heat source **118**. In some embodiments, the heat exchanger **116** may be a gas heat exchanger and may be coupled to a gas burner (e.g., a burner assembly) that combusts a fuel (e.g., air-fuel mixture), such as acetylene, natural gas, propane, another gas, or any combination thereof to produce combustion products having an elevated temperature that are directed into the heat exchanger **116**. When the air flow is directed across the heat exchanger **116**, the air flow may absorb heat from the combustion products, thereby increasing the temperature of the air flow. Thereafter, the air flow may then exit the packaged HVAC unit **100** at a higher temperature compared to the air flow entering the packaged HVAC unit **100**.

(45) During a cooling mode of the packaged HVAC unit **100**, the heat exchanger **116** may be inoperative (e.g., turned off). However, some of the combustion products generated during a previous heating mode may linger or remain within heat exchange tubes of the heat exchanger **116**. Additionally or alternatively, when the heat exchanger **116** is not operating, another flow of air (e.g., ambient air) may nevertheless flow or reside in the heat exchange tubes of the heat exchanger **116**. As a relatively cool air flow (e.g., supply air cooled by the evaporator **110**) is directed across the heat exchange tubes, air within the heat exchange tubes may lose heat to the relatively cool air flow, thereby causing any moisture within the air to condense and form liquid condensate within the heat exchange tubes of the heat exchanger **116**. To mitigate collection of the condensate within the heat exchange tubes, the heat exchanger **116** of the present disclosure is configured to enable removal of the liquid condensate from the heat exchange tubes while also mitigating contact between the condensate and other components of the heat exchanger **116** (e.g., the burner assembly). In this way, degradation, inefficiency, and/or other adverse effects that may otherwise be caused by the condensate is avoided. The features and aspects of the heat exchanger **116** are discussed in further detail below.

(46) To separate various components within the packaged HVAC unit **100**, the packaged HVAC unit **100** may include partitions **120** (e.g., panels, vestibule panels, dividers, separation plates, etc.). As an example, the partitions **120** may divide the internal volume defined by the housing **102** into a first volume **122**, which may contain the heat source **118** (e.g., burner assembly) of the heat exchanger **116**, a second volume **124** (e.g., supply air section) from the supply air flow may exit the packaged HVAC unit **100**, a third volume **126** that contains the condenser **108**, and a fourth volume **128** (e.g., return air section **104**) configured to receive air flow directed into the packaged HVAC unit **100**. Various components of the packaged HVAC unit **100** may also be oriented along a number of axes including a lateral axis **190**, a longitudinal axis **192**, and a vertical axis **194**.

(47) FIG. **6** is side view of an embodiment of a furnace **200** (e.g., heat exchanger) that can be used with or in any of the systems of FIGS. **1-5** or any other suitable HVAC system. For example, the furnace **200** of FIG. **6** may correspond to the heat exchanger **116** in FIG. **5**. The furnace **200** may be disposed within a housing **130** (e.g., support structure), such as a section of the housing **102** of FIG. **5**, a section of an air handler, a standalone housing, or any other suitable support structure. The housing **130** may include a first side **132** (e.g., top side, panel, etc.) and a base **134** (e.g., bottom side, panel, etc.). However, in some embodiments, the furnace **200** may not include the first side **132** and/or the base **134** of the housing **130**.

(48) A blower **140** (e.g., fan) may be coupled or secured to the first side **132** of the housing **130** and may be configured to generate or direct an air flow **500** along an air flow path **510** of the furnace **200**. The blower **140** may correspond to the blower **114** in FIG. **5**. The housing **130** may also include a vestibule panel **150** (e.g., side panel, panel, etc.), which may correspond to one of the

partitions **120** of FIG. **5**. In the embodiment illustrated in FIG. **6**, the furnace **200** includes a heat exchange section **202** coupled to the vestibule panel **150**. The heat exchange section **202** may include one or more heat exchange tubes **204**, with each heat exchange tube **204** having a first port **206** (e.g., first end, top end, upper end, inlet, upstream end, etc.) and a second port **208** (e.g., second end, bottom end, lower end, outlet, downstream end, etc.) that are each coupled to the vestibule panel **150**. The heat exchange tube **204** may extend from the first port **206** to the second port **208** in any suitable configuration, geometry, or arrangement. In the illustrated embodiment, the heat exchange tube **204** also includes a first bend **210** (e.g., top bend, upstream bend), a second bend **212** (e.g., middle bend, midstream bend), and a third bend **214** (e.g., bottom bend, downstream bend). The heat exchange tube **204** extends between each of the first port **206**, second port **208**, first bend **210**, second bend **212**, and third bend **214**. In this manner, the heat exchange tube **204** defines multiple passes (e.g., tube passes, tube segments, conduit segments, etc.) of the heat exchange tube **204** through which combustion products are directed and across which the air flow **500** is directed. More specifically, the heat exchange tube **204** defines a first pass **216** extending between the first port **206** and the first bend **210**, a second pass **218** extending between the first bend **210** and the second bend **212**, a third pass **220** extending between the second bend **212** and the third bend **214**, and a fourth pass **222** extending between the third bend **214** and the second port **208**. In some embodiments, one or more of the passes **216**, **218**, **220**, **222** may extend in a direction along the lateral axis **190** (e.g., in a horizontal direction, along a horizontal axis **272**). In other embodiments, one or more of the passes **216**, **218**, **220**, **22** may extend at an angle relative to the horizontal axis **272**, as described in greater detail below.

(49) The first port **206** may be coupled or secured to a first side **152** of the vestibule panel **150** proximate an inlet **160** (e.g., passage, hole, aperture, opening, channel) formed in the vestibule panel **150**, and the second port **208** may be coupled to the first side **152** of the vestibule panel **150** proximate an outlet **170** (e.g., passage, hole, aperture, opening, channel) formed in the vestibule panel **150**. The first and second ports **206**, **208** may be coupled to the inlet **160**, and outlet **170**, respectively, via a swedging process or technique (e.g., expanding the first port **206** of the heat exchange tube **204** with the first port **206** positioned within the inlet **160** of the vestibule panel **150**), welding, brazing, or any other mechanical fastening technique. Each of the passes **216**, **218**, **220**, and **222** may be configured to extend crosswise relative to a direction of the air flow **500** along the flow path **510**, as described in greater detail below. It should be understood that each of the features of the heat exchange tube **204** described above may be fluidly coupled to one another to enable flow of fluids (e.g., combustion products, liquid condensate) through the heat exchange tube **204** towards the outlet **170**, as described in greater detail below. Further, in some embodiments, the heat exchange section **202** may include one or more heat exchange tubes **204** having additional features, alternative features, fewer or more bends, fewer or more passes, and so forth, based on selected characteristics, implementations, and/or operating parameters of the furnace **200**. Further still, the heat exchange tubes **204** have different orientations (e.g., offset, aligned relative to one another) to facilitate various tube configurations that may reduce an overall size, height, and/or footprint of the furnace **200**.

(50) As discussed herein, the furnace **200** may also include a burner assembly **230** (e.g., combustor, heating element, burner system) configured to ignite a mixture of fuel and oxidant (e.g., air-fuel mixture) to generate combustion products. For example, the burner assembly **230** may be fluidly connected to a fuel source **232** and may also be fluidly coupled to the inlet **160** on a second side **154** of the vestibule panel **150**. The burner assembly **230** may include one or more burners (e.g., premix burners) configured to ignite the mixture of fuel and oxidant to generate the combustion products, which are then directed through the inlet **160** and into the first port **206** of the heat exchange tube **204** via the first port **206** fluidly coupled to the inlet **160**. That is, the burner assembly **230** and the first port **206** may be in fluid communication, such that the combustion products may generally travel from the burner assembly **230**, through the inlet **160**, through the

first port **206**, through the first, second, third, and fourth passes **216**, **218**, **220**, and **222**, and towards the second port **208** of the heat exchange tube **204**. The second port **208** of the heat exchange tube **204** may be fluidly coupled to the outlet **170**, thereby enabling the combustion products to pass through the second port **208** and into the outlet **170**.

(51) From the outlet **170**, the combustion products may be removed from the system (e.g., via an exhaust conduit). To this end, the furnace **200** may also include a draft inducer **240** (e.g., draft inducer blower, draft blower, draft fan, inducer fan) fluidly coupled to the outlet **170** on the second side **154** of the vestibule panel **150**. The draft inducer **240** is configured to facilitate flow of the combustion products through the heat exchange tube **204**. That is, the draft inducer **240** may be fluidly coupled to the second port **208** via the outlet **170** and may be configured to draw the combustion products through the heat exchange tube **204**. When operation of the furnace **200** is initiated to heat the air flow **500** (e.g., upon receipt of a call for heating), the draft inducer **240** may be operated prior to operation of the burner assembly **230** (e.g., 30 seconds before, a predetermined time period before, etc.), thereby removing any air or other gaseous compounds that may be present within the heat exchange tube **204** (e.g., via a suction air flow generated by the draft inducer **240**). The draft inducer **240** may also be coupled to an exhaust conduit (not shown) which may be configured to direct combustion gases, air, and/or other gaseous compound out of the furnace **200** (e.g., the HVAC system having the furnace **200**), as described in greater detail below.

(52) As discussed above, the burner assembly **230** may be coupled or secured to the vestibule panel **150** at the inlet **160** of the vestibule panel **150**, and the draft inducer **240** may be coupled or secured to the vestibule panel **150** at the outlet **170** of the vestibule panel **150**. The burner assembly **230** and the draft inducer **240** may be secured via fasteners, brackets, pins, screws, or any other suitable mechanical fastening technique. As illustrated, the inlet **160** is located above (e.g., vertically above) the outlet **170** relative to the base **134** of the furnace **200** (e.g., relative to gravity, relative to the vertical axis **194**, etc.). Thus, when installed and coupled to the vestibule panel **150**, the burner assembly **230** is located at a top portion **260** of the furnace **200**, and the draft inducer is located at a bottom portion **270** of the furnace **200**. That is, the burner assembly **230** is higher in position than the draft inducer **240** relative to the base **134** of the furnace **200** (e.g., relative to gravity, relative to the vertical axis **194**). This configuration (e.g., top-fired configuration, top-burner configuration) limits, reduces, and/or prevents the potential of liquid and/or liquid condensate that may form within the heat exchanger tube **204** from flowing toward the burner assembly **230**, as described in greater detail below.

(53) As previously described, operation of the furnace **200** may cause condensate to form within the heat exchange tube **204** as the air flow **500** travels across the heat exchange tube **204** along the flow path **510**, such as during a cooling mode of operation when the furnace **200** is not operating to heat the air flow **500**. As the liquid condensate forms within the heat exchange tube **204**, the liquid condensate may be directed away from the burner assembly **230** and towards the outlet **170**, such as via force of gravity. In some embodiments, each of the passes **216**, **218**, **220**, and **222** may generally extend along the lateral axis **190** and may be disposed at an angle relative to a horizontal axis **272** (e.g., a horizontal direction), such that condensate formed within the heat exchange tube **204** may directed away from the top portion **260** of the furnace **200** and towards the bottom portion **270** of the furnace **200** via gravity. The liquid condensate may flow through one or more of the passes **216**, **218**, **220**, and **222** and along one or more of the bends **210**, **212**, **214** towards the second port **208** of the heat exchange tube **204** that is in fluid communication with the outlet **170**. Liquid condensate that reaches the outlet **170** may then be discharged from the furnace **200** via a drain (e.g., drain outlet), a conduit, or any suitable discharge flow path fluidly coupled to the outlet **170**. In some embodiments, a gasket **180** (e.g., paper gasket) may be positioned between the outlet **170** of the vestibule panel **150** and the draft inducer **240**. The gasket **180** may surround the second port **208** of the heat exchange tube **204**, may have an opening formed therein that is aligned with the second port **208**, and may extend from the outlet **170** (e.g., in a horizontal direction along the

horizontal axis **272**) away from the vestibule panel **150**. The gasket **180** may be configured to facilitate drainage of the liquid condensate by providing clearance for the liquid condensate to drain out of the heat exchange tube **204** before reaching the draft inducer **240**. That is, the gasket **180** may be composed of a porous material, thereby enabling liquid condensate to drain through the gasket **180** and out of the furnace **200** before reaching the draft inducer **240**. It should be noted that various aspects of the furnace **200** may be manufactured, configured, and/or arranged to block or reduce an undesirable impact of the liquid condensate on the furnace **200** that may otherwise be caused by contact with the liquid condensate. By way of example, components of the heat exchange section **202**, such as the heat exchange tube **204**, the inlet **160**, the outlet **170**, the gasket **180**, and the vestibule panel **150** may be made of stainless steel, chromium, and/or other suitable (e.g., corrosion resistant) material to reduce undesirable effects of the liquid condensate on the structural integrity and/or performance of the components.

(54) The furnace **200** may also include a controller **250** configured to control operation of the burner assembly **230** and the draft inducer **240**, such as based on an operating mode of the furnace **200**. The controller **250** may be coupled to the vestibule panel **150** via welding, fasteners, screws, or other suitable technique. During operation, the controller **250** may receive a signal indicative of a call for operation in the cooling mode, and in response, the controller **250** may operate to shut-off or power down the burner assembly **230** and the draft inducer **240** such that combustion products are no longer generated and circulated through the heat exchange tubes **204**. At a different time, the controller may receive a signal indicative of a call for operation in the heating mode, and in response, the controller **250** may operate to activate or power on the draft inducer **240** and the burner assembly **230** (e.g., sequentially, power on the draft inducer **240** prior to powering on the burner assembly **230**, etc.) such that combustion products may be generated and circulated through the heat exchange tube **204** to enable heating of the air flow **500** directed across the heat exchange tube **204** along the air flow path **510**.

(55) In some circumstances, the controller **250** may be operate to activate the draft inducer **240** without activating the burner assembly **230**. For example, a presence of liquid condensate within the heat exchange tube **204** may be detected via one or more sensors **274** (e.g., a liquid sensor, humidity sensor, condensate sensor, etc. fluidly coupled to and/or disposed within the heat exchanger tube **204** and communicatively coupled to the controller **250**). Based on detection of the presence of liquid condensate, the draft inducer **240** may be activated to draw an air flow through the heat exchange tube **204** to motivate the liquid condensate towards the second port **208** and away from the burner assembly **230**. To facilitate control of the components of the furnace **200**, the controller **250** may include a memory **252** with instructions stored thereon for controlling operation the furnace **200** and components of the furnace **200**, and processing circuitry **254** configured to execute such instructions. For example, the processing circuitry **254** may include one or more application specific integrated circuits (ASICs), one or more field programmable gate arrays (FPGAs), one or more general purpose processors, or any combination thereof. Additionally, the memory **252** may include a non-transitory computer-readable medium that may include volatile memory, such as random-access memory (RAM), and/or non-volatile memory, such as read-only memory (ROM), optical drives, hard disc drives, optical drives, solid-state drives or any other suitable non-transitory computer-readable medium storing instructions that, when executed by the processing circuitry **254**, may control operation of the furnace **200**. Although FIG. **6** illustrates the controller **250** as being coupled to the vestibule panel **150**, in some embodiments, the controller **250** may be disposed elsewhere, such as remotely relative to the furnace **200**.

(56) FIG. **7** is a schematic side view of an embodiment of the furnace **200**, illustrating various flow directions (e.g., flow paths) of liquid (e.g., liquid condensate) that may form within the heat exchange tube **204**, such as in the manners described above. As illustrated, each of the passes **216**, **218**, **220**, **222** extends at least partially in a direction of the lateral axis **190** and generally crosswise to the air flow path **510** (e.g., crosswise to the vertical axis **194**) through which the air flow **500** is

directed across the heat exchange tube **204**. For example, one or more of the passes **216**, **218**, **220**, and **222** (e.g., passes **216**, **222**) of the heat exchange tube **204** may extend a length **620** (e.g., width, distance) from the vestibule panel **150**. However, some of the passes **216**, **218**, **220**, and **222** (e.g., passes **218**, **220**) may extend a length less than the length **620**. In some embodiments, the length **620** may be greater than a width **630** of the air flow path **510**, such that the air flow **500** directed along the air flow path **510** may contact each of the passes **216**, **218**, **220**, **222** as the air flow **500** flows through the air flow path **510**. When air (e.g., ambient air) within the heat exchange tube **204** is cooled via a relatively cool supply air flow (e.g., air flow **500**) directed along the air flow path **510** across the heat exchange tube **204**, such as during non-operation of the furnace **200**, moisture within the air inside the heat exchange tube **204** may condense, thereby forming liquid condensate within the heat exchange tube **204**. As mentioned above, one or more of the passes **216**, **218**, **220**, and **222** may be disposed at an angle relative to a horizontal axis **272** such that liquid condensate formed within the passes **216**, **218**, **220**, and **222** may be directed via gravity, through the heat exchange tube **204** towards the outlet **170** and the gasket **180**. Additionally, in some embodiments, one or more of the passes **216**, **218**, **220**, and **222** may extend in a direction along the lateral axis **190** (e.g., a horizontal direction, along the horizontal axis **272**).

(57) For example, the first port **206** may be secured to the inlet **160** (e.g., passage, hole, aperture, opening, channel) at a first position **300** (e.g., first location along the vertical axis **194**). The first pass **216** may extend from the first port **206** to the first bend **210** at a first angle **400** (e.g., downward angle) relative to the horizontal axis **272**, such that liquid condensate formed within the first port **206** and/or the first pass **216** may be directed along a first flow path **600** of the heat exchange tube **204** towards the first bend **210** via gravity. That is, the first bend **210** may be disposed at a second position **302** (e.g., second location along the vertical axis **194**), which is lower relative to gravity than the first position **300** of the first port **206**. Thus, condensate formed within the first port **206** and/or the first pass **216** may travel from the first position **300** to the second position **302** along the first flow path **600** via gravity. Upon reaching the first bend **210**, liquid condensate may fall (e.g., via gravity) along a second flow path **602** of the heat exchange tube **204** towards the second pass **218**. The second pass **218** may be fluidly coupled to the first bend **210** at a third position **304** (e.g., third location along the vertical axis **194**). As shown in the illustrated embodiment, the third position **304** is lower than the second position **302** relative to gravity such that condensate traveling through the first bend **210** falls along the second flow path **602** and into the second pass **218**.

(58) The second pass **218** may extend from the first bend **210** to the second bend **212** at a second angle **402** (e.g., downward angle) relative to the horizontal axis **272**, such that liquid condensate within the second pass **218** may be directed along a third flow path **604** of the heat exchange tube **204** towards the second bend **212** via gravity. That is, the second bend **212** may be disposed at a fourth position **306** (e.g., fourth location along the vertical axis **194**), which is lower relative to gravity than the third position **304**. Thus, condensate reaching the second pass **218** may travel from the third position **304** to the fourth position **306** along the third flow path **604** via gravity. Upon reaching the second bend **212**, liquid condensate may fall via gravity along a fourth flow path **606** of the heat exchange tube **204** towards the third pass **220**. The third pass **220** may be fluidly coupled to the second bend **212** at a fifth position **308** (e.g., fifth location along the vertical axis **194**). As shown in the illustrated embodiment, the fifth position **308** is lower than the fourth position **306** relative to gravity such that condensate traveling through the second bend **212** falls along the fourth flow path **606** and into the third pass **220**.

(59) The third pass **220** may extend from the second bend **212** to the third bend **214** at a third angle **404** (e.g., downward angle) relative to the horizontal axis **272**, such that liquid condensate within the third pass **220** may be directed along a fifth flow path **608** of the heat exchange tube **204** towards the third bend **214** via gravity. That is, the third bend **214** may be disposed at a sixth position **310** (e.g., sixth location along the vertical axis **194**) which is lower relative to gravity than

the fifth position **308**. Thus, condensate reaching the third pass **220** may travel from the fifth position **308** to the sixth position **310** along the fifth flow path **608** via gravity. Upon reaching the third bend **214**, liquid condensate may fall via gravity along a sixth flow path **610** of the heat exchange tube **204** towards the fourth pass **222**. The fourth pass **222** may be fluidly coupled to the third bend **214** at a seventh position **312** (e.g., seventh location along the vertical axis **194**). As shown in the illustrated embodiment, the seventh position **312** is lower than the sixth position **310** relative to gravity such that condensate traveling through the third bend **214** falls along the sixth flow path **610** and into the fourth pass **222**.

(60) The fourth pass **222** may extend from the third bend **214** to the second port **208** at a fourth angle **406** (e.g., downward angle) relative to the horizontal axis **272**, such that liquid condensate within the fourth pass **222** may be directed along a seventh flow path **612** of the heat exchange tube **204** towards the second port **208** via gravity. That is, the second port **208** may be disposed at an eighth position **314** (e.g., eight location along the vertical axis **194**) which is lower relative to gravity than the seventh position **312**. Thus, condensate reaching the fourth pass **222** may travel from the seventh position **312** to the eighth position **314** along the seventh flow path **612** via gravity. As discussed above, the embodiments included herein should not be considered limiting and other embodiments of the furnace **200** may include fewer or more passes, bends and heat exchange tubes as desired based on various design considerations of the furnace **200**. In the manner described above, the furnace **200** including the features described herein enables drainage and removal of liquid condensate from the furnace while also directing the liquid condensate away from the burner assembly **230**, thereby avoiding undesirable contact between liquid condensate and the burner assembly **230** and increasing efficiency and longevity of the burner assembly **230**.

(61) It should be noted that in some embodiments, one or more of the passes **216**, **218**, **220**, and **222** may not extend at an angle relative to the horizontal axis **272** and instead may generally extend in a direction along the lateral axis **190** (e.g., in a horizontal direction along the horizontal axis **272** as illustrated in FIG. **6**). That is, each heat exchange tube **204** may include one or more passes **216**, **218**, **220**, **222** that extend at an angle relative to the horizontal axis **272** across the flow path **510**, one or more passes **216**, **218**, **220**, **222** that extend along the horizontal axis **272** (e.g., in a horizontal direction) across the flow path **510**, or any combination thereof.

(62) FIG. **8** is a schematic side view of an embodiment of a portion of the furnace **200**, illustrating the draft inducer **240** and the gasket **180** configured to facilitate removal of liquid condensate from the furnace **200**. The gasket **180** may be disposed on the second side **154** of the vestibule panel **150** between the draft inducer **240** and the outlet **170** (e.g., passage, channel, hole, aperture, opening). As described above, liquid condensate that reaches the fourth pass **222** may travel along the seventh flow path **612** of the heat exchange tube **204** towards the second port **208**, the outlet **170**, and the gasket **180**. The gasket **180** may be disposed around (e.g., circumferentially around) the outlet **170** and around the port **208** and may extend to a drain outlet **282**. The gasket **180** may provide a channel, flow path, or other guide extending from the port **208**, through the outlet **170**, and to the drain outlet **282** such that liquid condensate directed along the seventh flow path **612** may flow from the outlet **170** and pass through or along the gasket **180** to be discharged from the furnace **200**. In some embodiments, the gasket **180** may be composed of a porous material, thereby enabling liquid condensate to pass through the gasket **180** and towards the drain outlet **282** to be discharged from the furnace **200**.

(63) During an operative mode (e.g., heating mode), the draft inducer **240** may be configured to discharge combustion products circulated through the heat exchange tube **204** via an exhaust outlet **280** (e.g., outlet port, discharge port), which may be fluidly coupled to the draft inducer **240**, such as via a panel (e.g., side panel) of the packaged HVAC unit **100** of FIG. **5**. In some embodiments, the exhaust outlet **280** may be fluidly coupled to a conduit **290** (e.g., vertical exhaust, exhaust conduit) configured to receive combustion products from the draft inducer **240** and direct flow of the combustion products in a direction **700** (e.g., vertical direction), as described in greater detail

below, to discharge the combustion products from the furnace **200** and/or the packaged HVAC unit **100**.

(64) FIG. **9** is a front perspective view of an embodiment of the furnace **200**, illustrating multiple heat exchange tubes **204** arranged along the longitudinal axis **192**. As illustrated, each of the heat exchange tubes **204** includes the first port **206**, which is fluidly coupled to the burner assembly **230** via respective inlets **160** (e.g., passage, channel, opening, aperture, hole) of the vestibule panel **150**, and may also include the second port **208**, which is fluidly coupled to the draft inducer **240** via respective outlets **170** (e.g., passage, channel, opening, aperture, hole) of the vestibule panel **150**. As noted above, the burner assembly **230** may be coupled to the vestibule panel **150** above the draft inducer **240** relative to gravity (e.g., along the vertical axis **194**). That is, the burner assembly **230** may be positioned above the draft inducer **240** such that the inlets **160** of the vestibule panel **150** are positioned above the outlets **170** of the vestibule panel **150** along the vertical axis **194**. Further, in some embodiments, a respective inlet **160** and the corresponding outlet **170** (e.g., inlet and outlet fluidly coupled together via a heat exchange tube **204**) are also aligned along the vertical axis **194** such that the first port **206** and the second port **208** of each respective heat exchange tube **204** are aligned with one another along the vertical axis **194**. For example, the burner assembly **230** may be coupled to the vestibule panel **150** such that a respective inlet **160** (e.g., a first inlet) is positioned a distance **800** from the base **134** of the housing **130** and a distance **808** from a side **136** of the housing **130**, and the draft inducer **240** may be coupled to the vestibule panel **150** such that a respective outlet **170** (e.g., a first outlet fluidly coupled to the first inlet **160** via a heat exchange tube **204**) is positioned a distance **802** from the base **134** of the housing **130** and a distance **810** from the side **136** of the housing **130**. The distance **800** may be greater than the distance **802**, and the distance **808** may be approximately equal to the distance **810**. Thus, each inlet **160** may be positioned within the vestibule panel **150** at a position above the corresponding outlet **170** relative to gravity such that the first port **206** of a respective heat exchange tube **204** is aligned with the corresponding second port **208** of the respective heat exchange tube **204** along the vertical axis.

(65) As discussed above, the furnace **200** may be part of an outdoor or rooftop HVAC unit. In some embodiments, the burner assembly **230** may also be positioned within a threshold distance **806** from the first side **132** (e.g., top side) of the housing **130**, thereby providing a desired clearance between the burner assembly **230** and the first side **132**. For example, during a heating mode, the burner assembly **230** may be operated to generate combustion gases to heat an air flow. By positioning the burner assembly **230** near the first side **132** (e.g., within a threshold distance **806** from the first side **132**), heat generated from the operation of the burner assembly **230** may melt snow accumulated on the first side **132** of the housing such that the snow may be directed away from the burner assembly **230** via gravity, thereby reducing undesirable effects on the structural integrity and/or performance of the components of the burner assembly **230** that may otherwise be caused by contact with water or other liquid.

(66) In some embodiments, the exhaust outlet **280** of the draft inducer **240** may be fluidly coupled to the conduit **290**, which may extend in the direction **700**, such as along the vertical axis **194**. As shown in the illustrated embodiment, the conduit **280** may extend in the direction **700** to a position above the first side **132** of the housing **130** (e.g., along the vertical axis **194**). Directing the combustion products along the exhaust conduit **280** in the direction **700** may also facilitate reducing undesirable effects on the structural integrity and/or performance of the components of the furnace **200**. For example, heat from the combustion products discharged via the conduit **290** may also be used to melt snow or other environmental conditions which may accumulate on the first side **132** of the housing **130** and may have an undesirable impact on the performance of the furnace **200** and/or may cause the furnace **200** to bear an undesired weight.

(67) FIG. **10** is front perspective view of an embodiment of the furnace **200**, illustrating multiple heat exchange tubes **204** arranged along the longitudinal axis **192**. As described above with respect to FIG. **9**, the respective inlets **160** may be positioned above the respective outlets **170** relative to

gravity, and thus the first port **206** of each heat exchange tube **204** may also be positioned above the respective second port **208** relative to gravity. In some embodiments, the heat exchange tubes **204** may also be arranged such that the first port **206** is offset (e.g., horizontally offset) from the corresponding second port **208** of the heat exchange tube **204** along the longitudinal axis **192**. For example, the burner assembly **230** may be coupled to the vestibule panel **150** such that a respective inlet **160** is positioned a distance **820** from the side **136** of the housing **130**, and the draft inducer **240** may be coupled to the vestibule panel **150** such that the corresponding outlet **170** (e.g., outlet fluidly coupled to the respective inlet via the heat exchange tube **204**) is positioned a distance **822** from the side **136** of the housing **130**. The distance **820** is greater than the distance **822**, such that the respective inlet **160** and the corresponding outlet **170** are offset (e.g., horizontally offset) from one another along the longitudinal axis **192** by a distance **824**. Accordingly, when installed, a respective heat exchange tube **204** may include a first port **206** that couples to the inlet **160** at the distance **820** from the side **136** of the housing **130**, and a second port **208** that couples to the outlet **170** at the distance **822** from the side **136** of the housing **130**, and thus, the first port **206** and the corresponding second port **208** of the respective heat exchange tube **204** may also be offset from one another along the longitudinal axis **192** by the distance **824**. In some embodiments, the distance **820** may be less than the distance **822**.

(68) In some embodiments, the respective inlets **160** and the corresponding outlets **170** may be offset from one another by the distance **824**, and the burner assembly **230** and the draft inducer **240** may nevertheless be aligned with one another along the vertical axis **194**. For example, the first port **206** of each heat exchange tube **204** may be fluidly coupled to a respective inlet **160**, the second port **208** may be fluidly coupled to a respective outlet **170**, and the heat exchange tubes **204** may each have a geometry or configuration that enables the first ports **206** and the corresponding second ports **208** to be offset from one another by the distance **824**. By arranging the inlets **160**, the outlets **170**, and the heat exchange tubes **204** in different orientations (e.g., inlet **160** and outlet **170** aligned with one another along the vertical axis **194**, inlet **160** and outlet **170** offset from one another relative to the longitudinal axis **194**, first and second ports **206**, **208** aligned with one another along the vertical axis **194**, first and second ports **206**, **208** offset from one another relative to the longitudinal axis **192**), an overall size, height, and/or footprint of the furnace **200** may be reduced, thereby reducing costs associated with manufacture, operation, and/or maintenance of the furnace **200**. For example, in the illustrated embodiment, the inlets **160** and corresponding outlets **170** are offset with one another relative to the longitudinal axis **192** (e.g., not aligned with one another along the vertical axis **194**), which reduces an overall height occupied by the furnace **200**.

(69) As described above with respect to FIG. 7, each of the heat exchange tubes **204** may include two or more passes (e.g., passes **216**, **218**, **220**, **222** of FIG. 7) and two or more bends (e.g., bends **210**, **212**, **214**). In some embodiments, one or more of the bends **210**, **212**, **214** may generally extend from one pass to another pass along the longitudinal axis **192** and may be disposed at an angle relative to the horizontal axis **272** (e.g., a horizontal direction) such that liquid condensate formed within the heat exchange tube **204** may be directed away from the burner assembly **230** via gravity. For example, the bend **212** may generally extend along the longitudinal axis **192** from the second pass **218** to the third pass **220** and may be disposed at an angle **408** relative to the horizontal axis **272** such that the bend **212** extends cross-wise to a direction of the airflow **500** (e.g., downward direction). In other embodiments, one or more of the bends **210**, **212**, **214** may extend in a direction along the vertical axis **194**. It should be noted that the heat exchange tubes **204** may each have a geometry or configuration that includes one or more bends that extend from one pass to another pass in a direction along the vertical axis **194**, one or more bends that extend from one pass to another pass in a direction along the longitudinal axis **192** at an angle relative to the horizontal axis **272**, one or more passes that extend in a direction along the lateral axis **190** (e.g., horizontal direction), one or more passes that extend in a direction along the lateral axis **190** at an angle relative to the horizontal axis **272**, or any combination thereof.

(70) As set forth above, the furnace of the present disclosure may provide one or more technical effects useful in the operation of HVAC systems, such as packaged HVAC units, configured to operate in a cooling mode and in a heating mode. For example, the furnace may be disposed within an air flow path of the HVAC system to enable the furnace to heat an air flow during operation of the furnace in the heating mode. During operation of the HVAC system in the cooling mode, relatively cool air may be directed across heat exchange tubes of the furnace, and air (e.g., ambient air) residing within the heat exchange tubes thereby be cooled. As a result, moisture within the air may condense and form liquid condensate within the heat exchange tubes. The top-fired burner assembly configuration disclosed herein enables discharge of the liquid condensate from the furnace while also mitigating contact between the liquid condensate and the burner assembly, thereby reducing adverse impacts on components of the HVAC system that may otherwise be caused by the liquid condensate. That is, the presently disclosed techniques may reduce a likelihood of wear and degradation to the HVAC system and its components that may be caused by water contact during operation of the HVAC system. The technical effects and technical problems in the specification are examples and are not limiting. It should be noted that the embodiments described in the specification may have other technical effects and can solve other technical problems.

(71) The techniques presented and claimed herein are referenced and applied to material objects and concrete examples of a practical nature that demonstrably improve the present technical field and, as such, are not abstract, intangible or purely theoretical. Further, if any claims appended to the end of this specification contain one or more elements designated as “means for [perform]ing [a function] . . . ” or “step for [perform]ing [a function] . . . ”, it is intended that such elements are to be interpreted under 35 U.S.C. 112(f). However, for any claims containing elements designated in any other manner, it is intended that such elements are not to be interpreted under 35 U.S.C. 112(f).

(72) While only certain features and embodiments of the disclosure have been illustrated and described, many modifications and changes may occur to those skilled in the art, such as variations in sizes, dimensions, structures, shapes and proportions of the various elements, values of parameters, including temperatures and pressures, mounting arrangements, use of materials, colors, orientations, and so forth without materially departing from the novel teachings and advantages of the subject matter recited in the claims. The order or sequence of any process or method steps may be varied or re-sequenced according to alternative embodiments. It is, therefore, to be understood that the appended claims are intended to cover all such modifications and changes as fall within the true spirit of the disclosure. Furthermore, in an effort to provide a concise description of the exemplary embodiments, all features of an actual implementation may not have been described, such as those unrelated to the presently contemplated best mode of carrying out the disclosure, or those unrelated to enabling the claimed disclosure. It should be noted that in the development of any such actual implementation, as in any engineering or design project, numerous implementation specific decisions may be made. Such a development effort might be complex and time consuming, but would nevertheless be a routine undertaking of design, fabrication, and manufacture for those of ordinary skill having the benefit of this disclosure, without undue experimentation.

Claims

1. A furnace for a heating, ventilation, and air conditioning (HVAC) unit, comprising: a heat exchange tube configured to flow combustion products therethrough and place the combustion products in a heat exchange relationship with an air flow directed across the heat exchange tube, wherein the heat exchange tube comprises a plurality of passes extending across a flow path of the air flow directed across the heat exchange tube; a burner assembly fluidly coupled to a first port of the heat exchange tube and configured to generate the combustion products directed into the heat exchange tube via the first port; a draft inducer blower fluidly coupled to a second port of the heat

exchange tube and configured to draw the combustion products through the heat exchange tube, wherein the burner assembly is higher in position than the draft inducer blower relative to a base of the HVAC unit in an installed configuration of the furnace with the HVAC unit; and a paper gasket fluidly coupled to and disposed circumferentially around the second port and configured to receive condensate formed during operation of the furnace, wherein a material of the paper gasket is porous and is configured to enable the condensate to drain through the material in a vertical direction away from the draft inducer blower.

2. The furnace of claim 1, comprising a panel, wherein the heat exchange tube is secured to the panel and is disposed on a first side of the panel, and the burner assembly and the draft inducer blower are coupled to a second side of the panel.

3. The furnace of claim 2, wherein the paper gasket is disposed between the panel and the draft inducer blower.

4. The furnace of claim 2, wherein the first port of the heat exchange tube fluidly couples with the burner assembly via a first passage through the panel and the second port of the heat exchange tube fluidly couples with the draft inducer blower via a second passage in the panel.

5. The furnace of claim 4, wherein the first passage is vertically above and horizontally offset from the second passage.

6. The furnace of claim 1, wherein the draft inducer blower is configured to discharge the combustion products from the HVAC unit via a side panel of the HVAC unit in the installed configuration of the furnace with the HVAC unit.

7. The furnace of claim 1, comprising a conduit configured to receive the combustion products from the draft inducer blower and direct flow of the combustion products in an additional vertical direction.

8. The furnace of claim 7, wherein the conduit is configured to discharge the combustion products from the HVAC unit in the installed configuration of the furnace with the HVAC unit.

9. A furnace for a heating, ventilation, and air conditioning (HVAC) system, comprising: a panel comprising an inlet and an outlet; a heat exchange tube fluidly coupled to the inlet and to the outlet on a first side of the panel and configured to direct combustion products from the inlet to the outlet and place the combustion products in a heat exchange relationship with an air flow directed across the heat exchange tube along an air flow path through the furnace, wherein the heat exchange tube comprises a plurality of tube passes extending across the air flow path; a burner assembly coupled to a second side of the panel at a first position along a vertical axis, wherein the burner assembly is configured to generate the combustion products and direct the combustion products into the heat exchange tube via the inlet; a draft inducer blower coupled to the second side of the panel at a second position along the vertical axis, wherein the draft inducer blower is configured to draw the combustion products through the heat exchange tube towards the outlet, wherein the first position is above the second position along the vertical axis; and a paper gasket fluidly coupled to and disposed circumferentially around the outlet, wherein the furnace is configured to direct condensate formed in the heat exchange tube along the paper gasket, and wherein the paper gasket is configured to enable the condensate to drain therethrough.

10. The furnace of claim 9, wherein the heat exchange tube comprises a first port fluidly coupled to the burner assembly via the inlet, and the heat exchange tube comprises a second port fluidly coupled to the draft inducer blower via the outlet.

11. The furnace of claim 10, wherein the first port and the second port are aligned with one another along the vertical axis.

12. The furnace of claim 10, wherein the first port and the second port are horizontally offset from one another relative to the vertical axis.

13. The furnace of claim 9, wherein the paper gasket is configured to direct the condensate in a direction along the vertical axis of the furnace toward a drain conduit.

14. A furnace for a heating, ventilation, and air conditioning (HVAC) system, comprising: a heat

exchange tube comprising a first port configured to receive combustion products and a second port configured to discharge the combustion products, wherein the heat exchange tube comprises a plurality of tube passes extending across a flow path of an air flow directed across the heat exchange tube, and wherein the heat exchange tube is configured to direct the combustion products from the first port to the second port; a burner assembly fluidly coupled to the first port, wherein the burner assembly is configured to generate the combustion products and direct the combustion products into the heat exchange tube via the first port; a draft inducer blower fluidly coupled to the second port, wherein the draft inducer blower is configured to draw the combustion products through the heat exchange tube and remove the combustion products from the heat exchange tube via the second port, wherein the first port is above the second port relative to gravity, and wherein the heat exchange tube is configured to discharge liquid condensate from the heat exchange tube via the second port; and a paper gasket fluidly coupled to and disposed circumferentially around the second port, wherein the furnace is configured to direct liquid condensate formed in the heat exchange tube along the paper gasket, and wherein a material composition of a material of the paper gasket is configured to enable the liquid condensate to pass through the material and out of the paper gasket.

15. The furnace of claim 14, comprising a panel comprising a first passage and a second passage formed therein, wherein the heat exchange tube is secured to the panel and is disposed on a first side of the panel, the burner assembly and the draft inducer blower are coupled to a second side of the panel, the first port is fluidly coupled to the burner assembly via the first passage, and the second port is fluidly coupled to the draft inducer blower via the second passage.

16. The furnace of claim 15, wherein the paper gasket is disposed on the second side of the panel between the second passage and the draft inducer blower, and the furnace comprises an exhaust conduit fluidly coupled to the draft inducer blower, wherein the exhaust conduit is configured to discharge the combustion products from the furnace.

17. The furnace of claim 1, wherein the plurality of passes is configured to direct the combustion products from the first port to the second port.

18. The furnace of claim 9, wherein each tube pass of the plurality of tube passes extends across the air flow path at a non-zero angle relative to a horizontal plane, and wherein the non-zero angle is a downward angle.

19. The furnace of claim 14, wherein the material composition of the material of the paper gasket is configured to enable the liquid condensate to drain through the paper gasket in a vertical direction away from the draft inducer blower.

20. The furnace of claim 1, wherein each pass of the plurality of passes extends across the flow path at a downward angle relative to a horizontal direction.
